

1) Reemplazar en la función original, lo específico para verificar si es una solución correcta

$$\frac{dz}{dx} = -(2Py_0 + Q)z = Pz^2$$

$$y = x$$

$$\frac{dy}{dx} = 1$$

$$\frac{dy}{dx} + \frac{y}{x} + \frac{y^2}{x^2} = 3$$

$$1 + \frac{x}{x} + \frac{x^2}{x^2} = 3$$

$$1 + 1 + 1 = 3$$

$$3 = 3$$

$$1 + \frac{dz}{dx} + \frac{x+z}{x} + \frac{(x+z)^2}{x^2} = 3$$

$$y = x+z$$

$$\frac{dy}{dx} = 1 + \frac{dz}{dx}$$

$$\frac{dz}{dx} + 1 + \frac{z}{x} + \frac{x^2 + 2xz + z^2}{x^2} = 3$$

$$\frac{dz}{dx} + 1 + \frac{z}{x} + 1 + \frac{2xz + z^2}{x^2} = 2$$

$$\frac{dz}{dx} + \frac{z}{x} + \frac{2z}{x^2} + \frac{z^2}{x^2} = 0$$

$$\frac{dz}{dx} + \frac{2z + z^2}{x^2} + \frac{z}{x} = 0$$

$$\frac{dz}{dx} = -\frac{2z + z^2 + zx}{x^2}$$

1) Verificar si $y = 1$ es solución particular de la ecuación $\frac{dy}{dx} + (2x-1)y - xy^2 = x-1$. En caso afirmativo, calcular la solución general.

$$Rta.: y = \frac{e^x(2-x)+c}{e^x(1-x)+c}$$

$$y = 1$$

$$\frac{dy}{dx} = 0$$

$$0 + (2x-1)1 - x = x-1$$

$$2x-1-x = x-1$$

$$x-1 = x-1$$

$$y = 1+z$$

$$\frac{dy}{dx} = \frac{dz}{dx}$$

$$\frac{dz}{dx} + (2x-1)(1+z) - x(1+z)^2 = x-1$$

$$\frac{dz}{dx} + 2x + 2xz - 1 - z - x(1+2z+z^2) = x-1$$

$$\frac{dz}{dx} + 2x + 2xz - 1 - z - x - 2xz - z^2x = x-1$$

$$\frac{dz}{dx} + x - z - z^2x = x$$

$$\frac{dz}{dx} = z^2x + z$$

$$-\left(\frac{1}{u}\right)^2 \frac{du}{dx} = \left(\frac{1}{u}\right)^2 x + \frac{1}{u}$$

$$\frac{du}{dx} = -x - u$$

$$\frac{du}{dx} + x + u = 0$$

$$u = y^{1-n}$$

$$u = z^{1-2}$$

$$u = z^{-1} \rightarrow z = u^{-1}$$

$$\frac{du}{dx} = -\frac{1}{z^2} \cdot \frac{dz}{dx}$$

$$\frac{du}{dx} \cdot -z^2 = \frac{dz}{dx}$$

$$y = e^{-\int P dx} \left[\int e^{\int P dx} \cdot Q dx + c \right]$$

$$P = 1$$

$$Q = 1$$

$$Q=x$$

$$y=e^{-\int dx}\left[\int e^{\int dx}\cdot x dx+c\right]$$

$$y=e^{-x}\int e^x\cdot x$$